Signals and Systems
SET214
Lecture (06)- Continuous Time systems‘ Types

Dr. Ahmed Reda Mohamed[1]

[1]The MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

Outlines

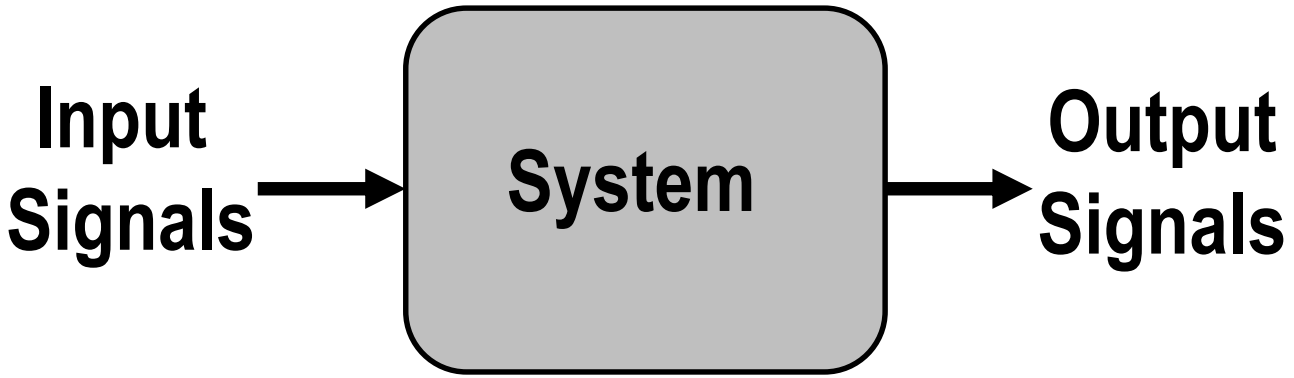
- Introduction to System
- Types of Systems
  - Causal and non-causal
  - Linear and Non-Linear
  - Time Variant and Time-invariant
  - Stable and Unstable
  - Static and Dynamic
  - Invertible and non-invertible Systems

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Introduction to System

- Systems process input signals to produce output signals.
- A system is a combination of elements that processes one or more signals to accomplish a function and produces output.



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Causal and non-causal

- **Causal** system:
  - A system is said to be causal if the present value of the output signal depends only on the present and/or past values of the input signal.

Ex:-

$y(t) = x(t)$

$y(t) = x(t-1)$

$y(t) = x(t) + x(t-1)$

- **Non-causal** system:
  - A system is said to be Non-causal if the present value of the output signal depends also on the future values of the input signal.

Ex:-

$y(t) = x(t+1)$

$y(t) = x(t) + x(t+1)$

$y(t) = x(t-1) + x(t+1)$

$y(t) = x(t-1) + x(t) + x(t+1)$

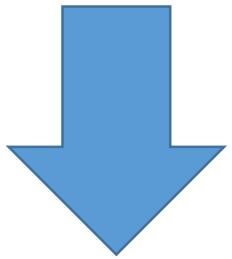
Exercise (1/4)

- Check whether the following are casual or non-casual system.

$y(t) = x(2t)$

Ans.

- Substitute $t=1$ in the above then $y(1) = x(2)$. Hence the given System is Non-Casual
- Substitute $t=-1$ in the above then $y(-1) = x(-2)$ Hence the given System is Casual

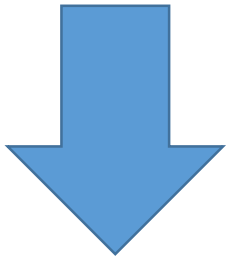


System is Non-Casual

$y(t) = x(-t)$

Ans.

- Substitute $t=1$ in the above then $y(1) = x(-1)$. Hence the given System is System is casual.
- Substitute $t=-1$ in the above then $y(-1) = x(1)$. Hence the given System is System is Non-casual.



System is Non-Casual

Exercise (2/4)

❑ Check whether the following are casual or non-casual system.

$y(t) = x(\sin(t))$

Ans.

❑ Substitute $t = \pi$ in the above $y(\pi) = x(0)$ ($\pi = 3.14$), System is Casual.

❑ Substitute $t = -\pi$ in the above $y(-\pi) = x(0)$ ($-\pi = -3.14$), System is Non-casual.

System is Non-Casual

$y(t) = \sin(t+2)\, x(t-1)$

Constant coefficient — Past value

Ans.

❑ Substitute $t = 1$ in the above $y(1) = \sin(3)\, x(0)$, System is Casual.

❑ Substitute $t = -1$ in the above $y(-1) = \sin(1)\, x(-2)$, System is Casual.

Constant coefficient — Past value

System is Casual

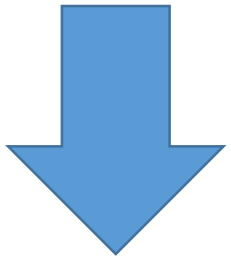
Exercise (3/4)

- Check whether the following are casual or non-casual system.

$$y(t) = \begin{cases} x(2t), & t < 0 \\ x(t-1), & t \geq 0 \end{cases}$$

Ans.

- Substitute $t = -1$ in the above eq. $y(-1) = x(-2)$ System is Casual.
- Substitute $t = 1$ in the bottom eq. $y(1) = x(0)$ System is Casual.



System is Casual

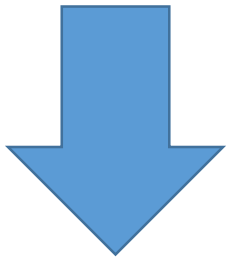
Exercise (4/4)

❑ Check whether the following are casual or non-casual system.

$$y(t) = \int_{-\infty}^{t} x(t)\,dt$$

Ans.

❑ Present output depends on present and past values. Hence, the system is casual

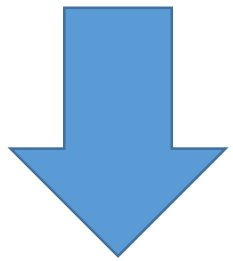


System is Casual

$$y(t) = \int_{-\infty}^{t+1} x(t)\,dt$$

Ans.

❑ Present output depends on future values Hence, the system is Non-casual



System is Non-casual

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Linear and Non Linear Systems (1/2)

- A system is said to be linear if it satisfies the principle of superposition or if it satisfies the properties of Additivity and Homogeneity (scaling).

- Consider a system where an input of $x_1(t)$ produces an output of $y_1(t)$. Further suppose that a different input, $x_2(t)$ produces another output, $y_2(t)$. The system is said to be additive, if an input of $x_1(t) + x_2(t)$ results in an output of $y_1(t) + y_2(t)$, for all possible input signals.

$$x_1(t) \rightarrow y_1(t) \quad \& \quad x_2(t) \rightarrow y_2(t) \Rightarrow x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$$

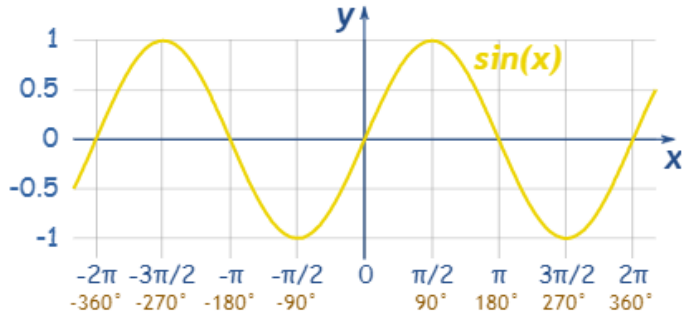
- Homogeneity (Scaling) means that a change in the input signal's amplitude results in a corresponding change in the output signal's amplitude. In mathematical terms, if an input signal of x(t) results in an output signal of y(t), an input of cx(t) results in an output of cy(t), for any input signal and c is a constant.

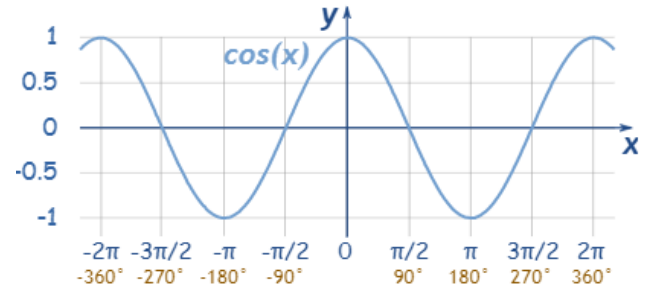
$$x(t) \rightarrow y(t) \Rightarrow Cx(t) \rightarrow Cy(t)$$

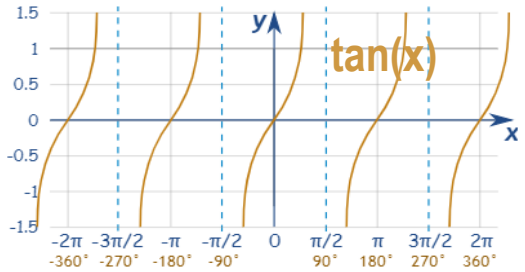
$$ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$ Super position

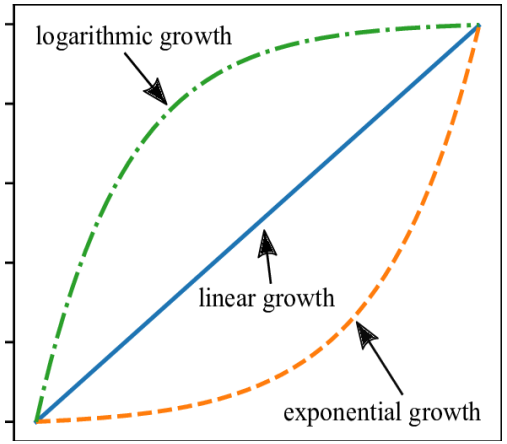
Linear and Non Linear Systems (2/2)

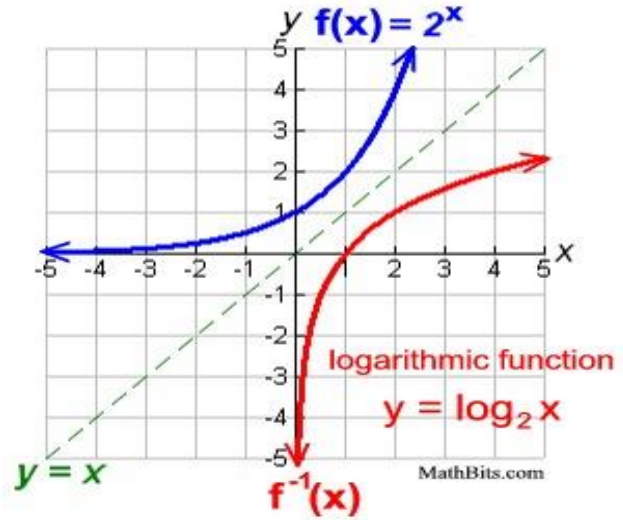
- For linearity System
  - Output should be zero for zero input.
  - It should not contain any nonlinear operation/function.
    - Example : The functions like Sin, Cos, tan, Cot, Sec, cosec, Log, Exponential, Modulus, Square, Cube, Root, Sampling function(), sinc(), Sgn() etc..... have nonlinear operations.











Exercise (1/4)

❑ Determine whether or not each of the following systems are linear with input x(t) and output y(t)

$y(t)=ax(t)+b$

Ans

❑ Step (1):-

$y_1(t)=ax_1(t)+b$ and ,$y_2(t)=ax_2(t)+b$

$y_1(t)+y_2(t)=ax_1(t)+b+ ax_2(t)+b$

$y_1(t)+y_2(t)=ax_1(t)+ax_2(t)+2b$

❑ Step (2):-

$y(t)=a(x_1(t)+x_2(t))+b$

$y(t)=ax_1(t)+ax_2(t)+b$

Check

$y_1(t)+y_2(t)\neq y(t)$

$ax_1(t)+ax_2(t)+2b \neq ax_1(t)+ax_2(t)+b$

System is non-linear

Exercise (2/4)

❑ Determine whether or not each of the following systems are linear with input x(t) and output y(t)

$y(t)=t.x(t)$

Ans

❑ Step (1):-

$y_1(t)=t.x_1(t)$ and $,y_2(t)=t.x_2(t)$

$y_1(t)+y_2(t)=t.x_1(t)+ t.x_2(t)$

❑ Step (2):-

$y(t)=t.(x_1(t)+x_2(t))$

$y(t)=t.x_1(t)+t.x_2(t)$

Check

$y_1(t)+y_2(t)=y(t)$

$t.x_1(t)+t.x_2(t) = t.x_1(t)+t.x_2(t)$

System is linear

Exercise (3/4)

❑ Determine whether or not each of the following systems are linear with input x(t) and output y(t)

$$y(t)=x(t).x(t-1)$$

Ans

❑ Step (1):-

$y_1(t)=x_1(t).x_1(t-1)$ and $y_2(t)= x_2(t).x_2(t-1)$

$y_1(t)+y_2(t)= x_1(t).x_1(t-1) + x_2(t).x_2(t-1)$

❑ Step (2):-

$y(t)=(x_1(t)+x_2(t)).(x_1(t-1)+x_2(t-1))$

$y(t)= x_1(t).x_1(t-1)+ x_1(t).x_2(t-1)+ x_2(t).x_1(t-1)+ x_2(t).x_2(t-1)$

Check

$$y_1(t)+y_2(t)\neq y(t)$$

$$x_1(t).x_1(t-1) + x_2(t).x_2(t-1) \neq x_1(t).x_1(t-1)+ x_1(t).x_2(t-1)+x_2(t).x_1(t-1)+ x_2(t).x_2(t-1)$$

System is non-linear

Exercise (4/4)

❑ Determine whether or not each of the following systems are linear with input x(t) and output y(t)

$y(t)=[x(t)]^2$

$y(t)=5x(t)$

$y(t)=3x(t)+2$

$y(t)=\cos(x(t))$

$y(t)=x(\cos(t))$

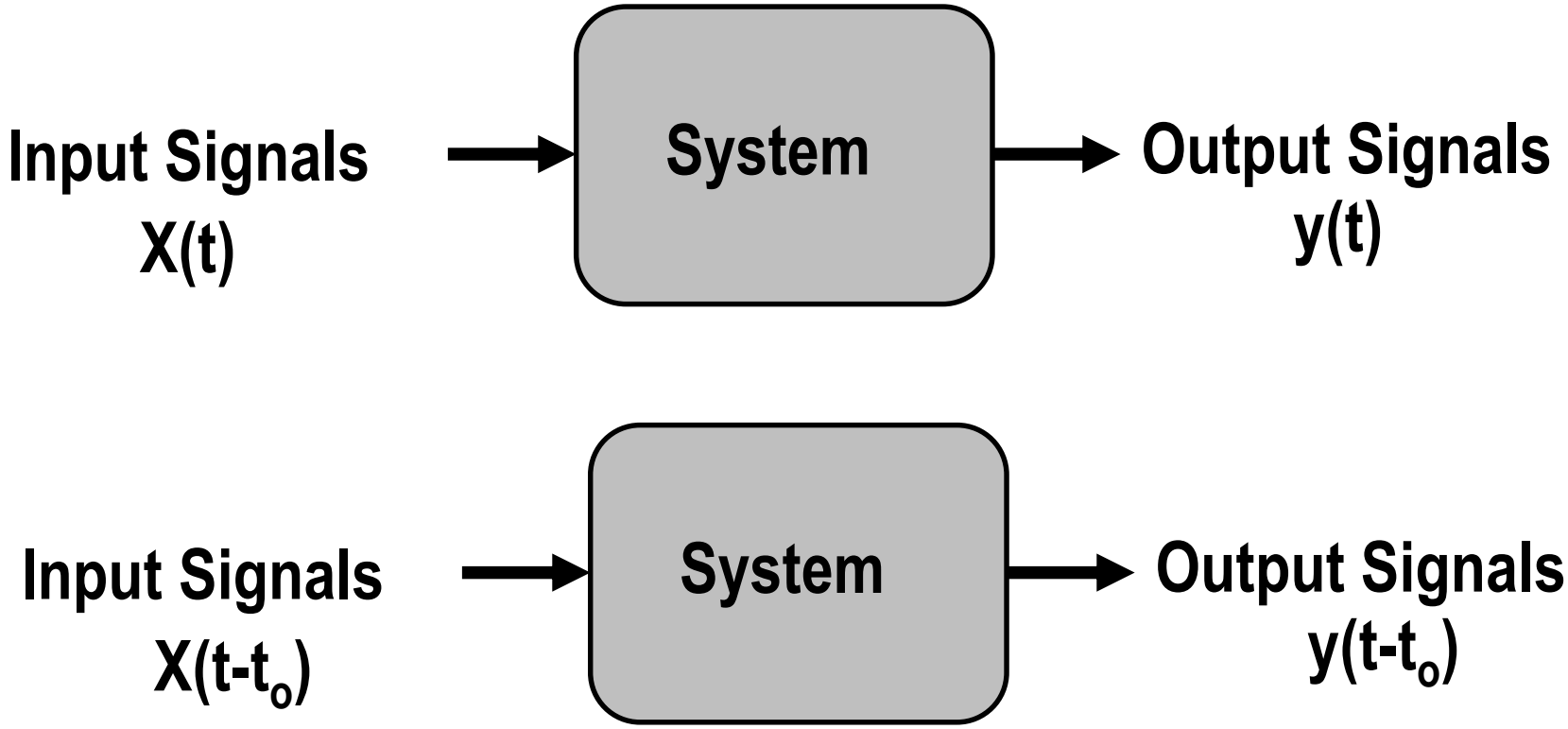
Ans

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Time Invariant/Variant Systems

- A system is said to be time invariant if a time delay or time advance of the input signal leads to a identical time shift in the output signal.
- A system is called time-invariant if a time shift in the input signal $x(t-t_o)$ causes the same time shift in the output signal $y(t-t_o)$.



Time- Invariant System

Exercise (1/4)

- Determine whether or not each of the following systems are time-invariant with input x(t) and output y(t)

$$y(t)=t.x(t)$$

Ans

- **Step (1):-**

$$y_1(t)= t.\ x(t-t_o)$$

Produce shift in the input

- **Step (2):-**

$$y(t-t_o)=(t-t_o).x(t-t_o)$$

Produce shift in the output

Check

$$y_1(t)\neq y(t-t_o)$$

$$t.\ x(t-t_o) \neq (t-to).x(t-to)$$

System is not time-invariant

System is time-variant

Exercise (2/4)

- Determine whether or not each of the following systems are time-invariant with input x(t) and output y(t)

$$y(t)=2+x(t)$$

Ans

- **Step (1):-**

$$y_1(t)= 2+x(t-t_o)$$

Produce shift in the input

- **Step (2):-**

$$y(t-t_o)=2+x(t-t_o)$$

Produce shift in the output

Check

$$y_1(t)=y(t-t_o)$$

$$2+x(t-t_o)=2+x(t-to)$$

System is time-invariant

Exercise (3/4)

- Determine whether or not each of the following systems are time-invariant with input x(t) and output y(t)

$$y(t)=\cos(t).x(t)$$

Ans

- **Step (1):-**

$$y_1(t)= \cos(t).x(t-t_o)$$

Produce shift in the input

- **Step (2):-**

$$y(t-t_o)=\cos(t-t_o).x(t-t_o)$$

Produce shift in the output

Check

$$y_1(t)\neq y(t-t_o)$$

$$\cos(t).x(t-t_o) \neq \cos(t-t_o).x(t-t_o)$$

System is time-variant

Exercise (4/4)

- Determine whether or not each of the following systems are time-invariant with input x(t) and output y(t)

$y(t)=3x(t)$

$y(t)=5\cos(x(t))$

$y(t)=5\cos(t)\,x(t)$

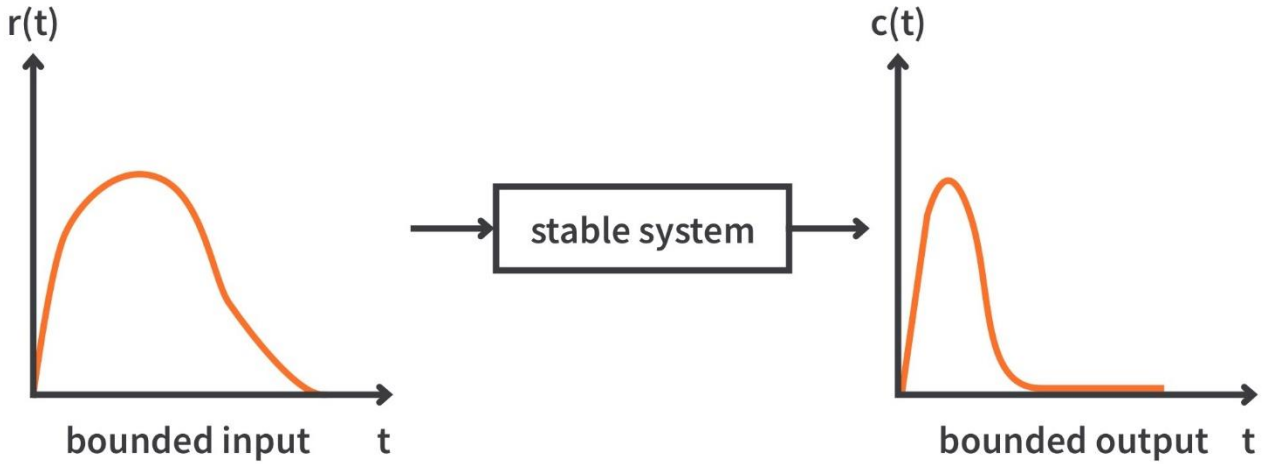
Ans

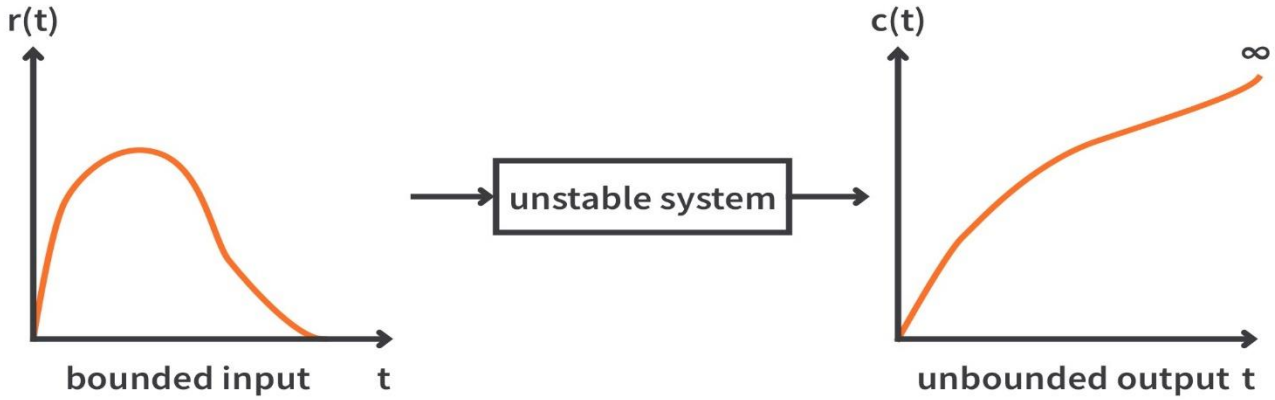
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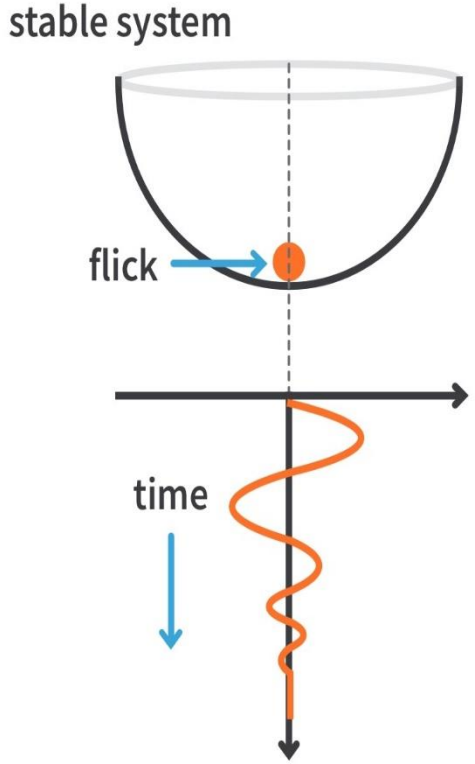
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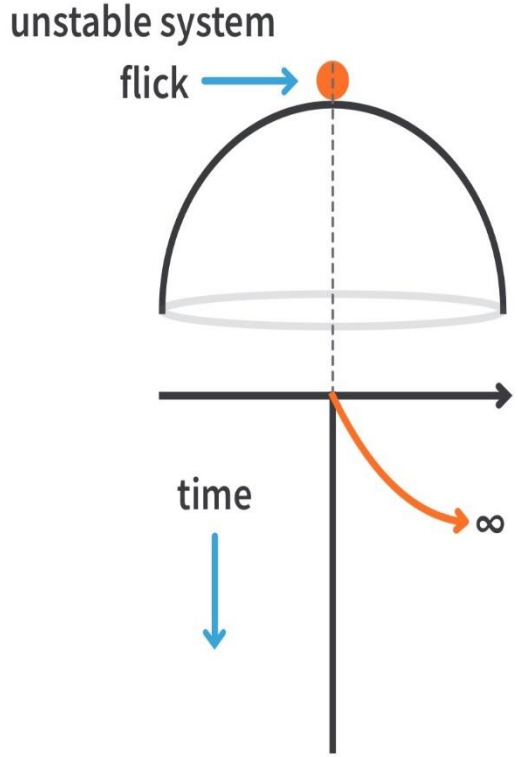
Stable/Unstable Systems

- A system is said to be bounded-input bounded- output stable (BIBO stable) if every bounded input results in a bounded output.









Exercise

❑ Determine whether or not each of the following systems are stable with input x(t) and output y(t)

$y(t)=x(t)+3$

Ans

Let x(t)=2 → y(t)=2+3=5

As input is bounded value, output is also a bounded value. Hence System is Stable

$y(t)=t.x(t)$

Ans

Let x(t)=2 → y(t)=2.t

As input is bounded value, output is not bounded value. Hence System is Un-Stable

$y(t)=[x(t)]^2$

Ans

Let x(t)=2 → $y(t)=2^2=4$

As input is bounded value, output is also a bounded value. Hence System is Stable

$y(t)=\frac{x(t)}{\sin(t)}$

Ans

Let x(t)=2 → $y(t)=\frac{2}{\sin(t)}$

As input is bounded value, output is not bounded value. Hence System is Un-Stable

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Static/Dynamic Systems

- A static system is memory less system
- It has no storage devices
- Its output signal depends on present values of the input signal

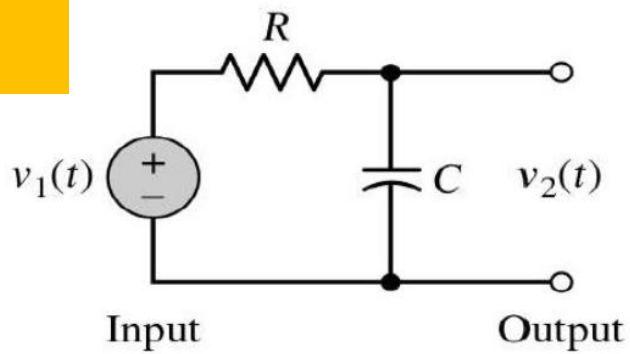
$$V(t)=R.\ I(t)$$

- A dynamic system possesses memory
- It has the storage devices
- A system is said to possess memory if its output signal depends on past values and future values of the input signal

$$v(t) = \frac{1}{C}\int_{t0}^{t1} i(t)\,dt$$

$$y(t) = x(t) - x(t - to)$$

Static or Dynamic?



- The system shown above is RC circuit
  - R is memoryless
  - C is memory device as it stores charge because of which voltage across it can't change immediately
- Hence given system is dynamic or memory system

Exercise

❑ Check whether given system is Static (memory-less) or Dynamic (memory).

Ans.

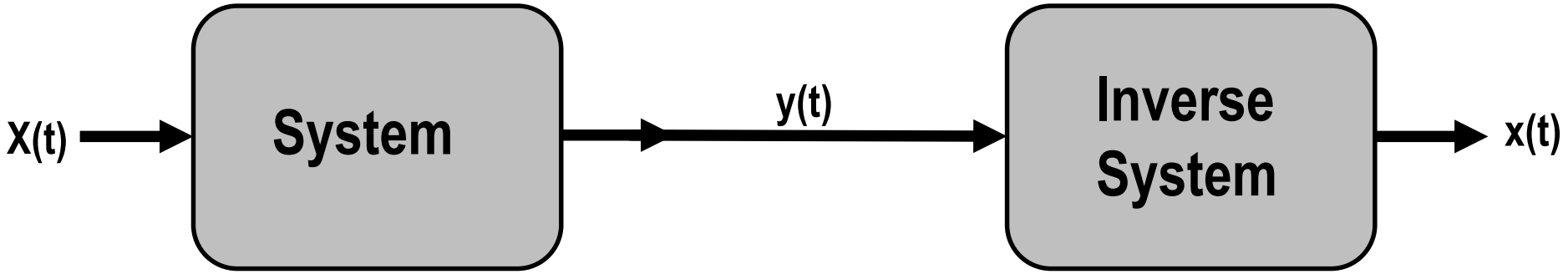
$y(t) = x(t) + x(t-1)$	Dynamic
$y(t) = x(-t)$	Dynamic
$y(t) = x(\sin t)$	Dynamic
$y(t) = x(t-1)$	Dynamic

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Invertible/Non-invertible Systems

❑ If a system is invertible if it has an Inverse System. Otherwise it is non-invertible system.



Example: y(t)=2x(t)

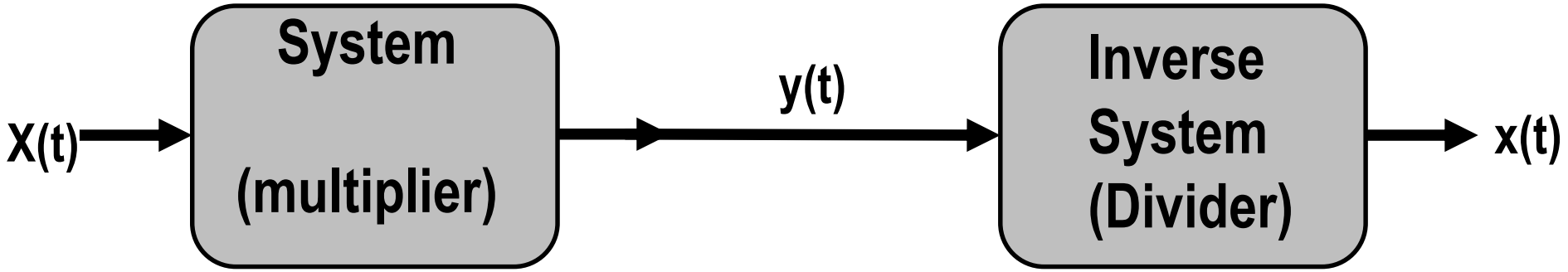
– System is invertible → must have inverse, that is:

– For any x(t) we get a distinct output y(t)

– Thus, the system must have an Inverse

x(t)=1/2 y(t)

Different inputs leads to different output→ system is invertible



Exercise (1/3)

- Determine whether or not each of the following systems are invertible with input x(t) and output y(t)

$$y(t)= x(t)+10$$

Ans

- **Step (1):-**

Set $x_1(t)=1$

$y_1(t)=1+10=11$

- **Step (2):-**

Set $x_2(t)=-1$

$y_2(t)=-1+10=9$

Different inputs leads to different output→ system is invertible

Check **System is invertible**

Exercise (2/3)

- Determine whether or not each of the following systems are invertible with input x(t) and output y(t)

$$y(t)= [x(t)]^2$$

Ans

- Step (1):-

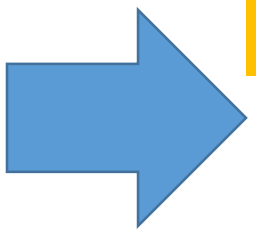
Set $x_1(t)=1$

$y_1(t)=1$

- Step (2):-

Set $x_2(t)=-1$

$y_2(t)=1$



Different inputs leads to same output→ system is invertible

Check System is non invertible

Exercise (3/3)

- Determine whether or not each of the following systems are invertible with input x(t) and output y(t)

$$y(t)= \sin (x(t))$$

Ans

- Step (1):-

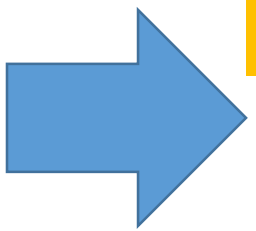
Set $x_1(t)=\pi$

$y_1(t)=0$

- Step (2):-

Set $x_2(t)=-\pi$

$y_2(t)=0$



Different inputs leads to same output→ system is invertible

Check System is non invertible

Thanks!